Dr. Christopher Rose
Associate Professor
Associate Director, WINLAB
Electrical and Computer Engineering
Rutgers University
73 Brett Road
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Education

Ph.D. in EECS, 1985, M.I.T., Cambridge, MA

S.M. in EECS, 1981, M.I.T., Cambridge, MA

B.S. in EECS, 1979. M.I.T., Cambridge, MA

Employment

1999–: *Associate Director*, WINLAB, Rutgers University

1996–: *Associate Professor*, Electrical and Computer Engineering, Rutgers University

1990–1996: *Assistant Professor*, Electrical and Computer Engineering, Rutgers University

1994–1999: *Director of Mobility Research*, WINLAB, Rutgers University

1985–1990: *Member Technical Staff*, AT&T Bell Laboratories, Network Systems Research Department

Awards

Outstanding Service as General Chair, ACM SIGMOBILE MobiCom 2001

Outstanding Service as Executive Committee Member, 1997-2001, ACM SIGMOBILE

Henry Rutgers Research Fellow, 1990

AT&T Bell Laboratories Cooperative Research Fellowship, 1979–1985

NSF Fellowship, 1979 (declined)

GEM Fellowship, 1979 (declined)

National Achievement 4-year Scholarship, 1975–1979

Current Research

Interference Avoidance, Mobility management, Communications Theory, New Wireless Architectures (infostations), Peaceful Coexistence in unlicensed national information infrastructure (U-NII) – 5GHz bands

Publications

Interference Avoidance, D. Popescu & C. Rose, Wiley Encylopedia of Telecommunications, J. Proakis, Ed., Wiley, 2002

Paging and Registration in Mobile Networks, C. Rose, Wiley Encylopedia of Telecommunications, J. Proakis, Ed., Wiley, 2002

Wireless Systems and Interference Avoidance, C. Rose, S. Ulukus, & R. Yates, IEEE Transactions on Wireless Communications (to appear July, 2002).

CDMA Codeword Optimization: interference avoidance and convergence via class warfare, IEEE Trans. Information Theory, 47(6), pp. 2368–2382, September (2001).

Infostations: New Perspectives On Wireless Data Networks, in **Next Generation Wireless Networks**, Editor: S. Tekinay, Kluwer Academic Publishers, May 2000.

One Dimensional Location Area Design, C.U. Saraydar, O.E. Kelly & C. Rose, IEEE Trans. Vehic. Tech 49(5) pp.1626–1632, September (2000).

State-Based Paging/Registration: A Greedy Approach, C. Rose, IEEE Transactions on Vehicular Technology 48(1), p166–173, January (1999).

Highly Mobile Users and Paging: optimal polling strategies, A. Yener & C. Rose, IEEE Transaction on Vehicular Technology, 47(4), pp.1251-1257, November (1998).

Location Uncertainty in Mobile Networks: a theoretical framework, C. Rose & R. Yates IEEE Communications Magazine, 35(2), pp.94–101, Februrary (1997).

Ensemble Polling Strategies for Increased Paging Capacity in Mobile Communication Networks, C. Rose & R. Yates, ACM Wireless Networks, 3(2), pp.159–167 (1997).

Genetic Algorithms Applied to the Cellular Call Admission Problem: Local Policies, A. Yener & C. Rose, IEEE Transactions on Vehicular Technology, 46(1), February (1997)

Analysis of a Mobile Assisted Adaptive Location Management Strategy, R. Yates, C. Rose, S. Rajagopalan & B. Badrinath, ACM Mobile Networks and Applications (MONET), 1(2), pp.105–112 (1996)

Minimizing the Average Cost of Paging and Registration: a timer-based method, C. Rose, ACM Wireless Networks, 2(2) pp.109–116, June (1996).

Genetic Algorithms and Call Admission to Telecommunications Networks, C. Rose & R. Yates, Computers and Operations Research, 23(5), pp.485–499, May (1996).

Scheduling Arrivals to Queues for Minimum Average Blocking: the S(n)/M/C/C system, C. Rose & R. Yates, Computers and Operations Research, 22(8), pp.793–806, October (1995).

Minimizing the Average Cost of Paging Under Delay Constraints, C. Rose & R. Yates, ACM Wireless Networks, 1(2) pp.211–219, (1995).

Mean Internodal Distance in Multihop Store & Forward Networks, C. Rose, IEEE Transactions on Communications, 40(8), pp.1310-1318, (1992).

Low Mean Internodal Network Topologies and Simulated Annealing, C. Rose, IEEE Transactions on Communications, 40(8), pp.1319-1326, (1992).

High T_c Superconductor Waveguides: Theory and Application, J.H. Winters & C. Rose, IEEE Transactions on Microwave Theory and Techniques, 39(4), pp.617-623, (1991).

A Dielectric-Free Superconducting Coaxial Cable, C. Rose & M.J. Gans, IEEE Transactions on Microwave Theory and Techniques, Vol. 38(2), pp.166-177, (1990).

Rapid Optimal Scheduling for Time-Multiplex Switches Using a Cellular Automaton, C. Rose, IEEE Transactions on Communications, 37(5), pp.500-509, (1989).

Minimum Distance Automata in Parallel Networks for Optimum Classification, J.H. Winters & C. Rose, Neural Networks, v.2, pp.127-132, (1989).

Frequency Dependence of Synchronization of Cochlear Nerve Fibers in the Alligator Lizard: evidence for a cochlear origin of timing and non-timing neural pathways, C. Rose & T.F. Weiss, Hearing Research, 33, pp.151-166, (1988).

Stages of Degradation of Timing Information in the Cochlea: a comparison of hair-cell and nerve-fiber responses in the alligator lizard, T.F. Weiss & C. Rose, Hearing Research, 33, pp.167-174, (1988).

A Comparison of Synchronization Filters in Different Auditory Receptor Organs, T.F. Weiss & C. Rose, Hearing Research, 33, pp.175-179, (1988).

The Performance of Random and Optimal Scheduling in a Time-Multiplex Switch, C. Rose & M.G. Hluchyj, IEEE Transactions on Communications, 35(8), pp.813-817, (1987).

Conference Publications

Interference Avoidance and Multiaccess Vector Channels, D.C. Popescu, O. Popescu and C. Rose, International Symposium on Information Theory (ISIT 2002), July 2002, Lusanne, Switzerland

Codeword Optimization for Asynchronous CDMA Systems Through Interference Avoidance, D.C. Popescu and C. Rose, 36th Conf. on Information Sciences and Systems (CISS 2002), March 20-22 2002, Princeton University, New Jersey.

Interference Avoidance and Multiaccess Dispersive Channels, D. C. Popescu and C. Rose, 35th Annual Asilomar Conference on Signals, Systems, and Computers, November 4-7 2001, Pacific Grove, California.

Fading Channels and Interference Avoidance, D. C. Popescu and C. Rose, 39th Allerton Conference on Communication, Control, and Computing, October 3-5 2001, Monticello, Illinois.

Minimizing Total Square Correlation with Multiple Receivers, O. Popescu and C. Rose, 39th Allerton Conference on Communication, Control, and Computing, October 3-5 2001, Monticello, Illinois.

Interference Avoidance and Sum Capacity for Multibase Systems O. Popescu and C. Rose, 39th Allerton Conference on Communication, Control, and Computing, October 3-5 2001, Monticello, Illinois.

A New Approach to Multiple Antenna Systems, CISS'01 March 2001, Baltimore, MD.

Wireless Data: patience has its rewards, C. Rose, IEEE CCW, October 18, 2000, Isla Captiva FL

MINE MINE MINE: Information Theory, Infostation Networks and Resource Sharing, A. Iacono and C. Rose, WCNC 2000, September 2000, Chicago.

Codeword Quantization for Interference Avoidance, D.C. Popescu and C. Rose, ICASSP 2000, June 2000, Istanbul.

Infostations: New Perspectives On Wireless Data Networks, A. Iacono and C. Rose, NJIT Symposium on Next Generation Wireless Networks, May 2000

Interference Avoidance for Wireless Systems, C. Rose, S. Ulukus, and R. Yates, IEEE Vehicular Technology Conference 2000, May 2000, Tokyo

Bounds on File Delivery Delay in an Infostations System, A. Iacono and C. Rose, IEEE Vehicular Technology Conference 2000, May 2000, Tokyo.

Interference Avoidance, Sum Capacity and Class Warfare, C. Rose, CISS 2000, March 2000, Princeton.

Interference Avoidance and Dispersive Channels: a new look at multicarrier modulation, D. Popescu and C. Rose, 37th Annual Allerton Conference on Communication, Control, and Computing, Allerton House, Monticello, Illinois, Sept. 1999.

The Use of Extended Kalman Filter for Emitter Localization in a Multipath Environment, D. Popescu and C. Rose, pp147–150, CISS'99, Baltimore.

Wireless Subscriber Mobility Management using Adaptive Individual Location Areas for PCS Systems, Z. Lei and C. Rose, ICC'98.

Location Area Design Using Population and Traffic Data, C. Saraydar and C. Rose, CISS'98, Princeton.

A Probability Criterion Based Location Tracking Method, Z. Lei and C. Rose, Proceedings of Globecom 97, Nov 2-8 1997, Phoenix AZ v. 2, pp. 977-981

Optimum Power Scheduling for CDMA Access Channels, A. Yener, C. Rose and R.D. Yates, Proceedings of Globecom 97, Nov 2-8 1997, Phoenix AZ v.3, pp. 1499-1503

Location Uncertainty in Mobile Networks: a theoretical framework, Christopher Rose & Roy Yates, IEEE/VTC'97, Phoenix, pp.597–601 April (1997).

Soft Dropping Power Control, R. Yates, S. Gupta, C. Rose & S. Sohn, IEEE/VTC'97, Phoenix, pp.1694–1698, April (1997).

Minimizing the Paging Channel Bandwidth for Cellular Traffic, C. Saraydar & C. Rose, ICUPC'96, Boston, pp. 941–945, October (1996) MA.

A Greedy Method of State-Based Registration, C. Rose, ICC'96, Dallas, pp. 1158–1162, June (1996)

Ensemble Polling Strategies for Mobile Communication Networks, C. Rose & R. Yates, IEEE VTC'96, Atlanta, pp. 101–105, April (1996).

Paging Strategies for Highly Mobile Users, A. Yener & C. Rose, pp. 1839–1842, IEEE VTC'96, Atlanta, GA, 4/96.

Minimization of Paging and Registration Costs Through Registration Deadlines, C. Rose, IEEE ICC'95, Seattle, pp.735–739, June, 1995.

Packet Arrival Scheduling at an Exponential Server for Minmax Blocking, R. Yates & C. Rose, IEEE ICC'95, Seattle, pp.798–802, June, 1995.

Paging Cost Minimization Under Delay Constraints, C. Rose & R. Yates, IEEE Infocom'95, Boston, pp.490–495, April 1995.

Local Call Admission Policies for Cellular Networks Using Genetic Algorithms, A. Yener & C. Rose, Conference on Information Science and Systems (CISS'95), Baltimore, pp.25–30, March 1995.

Near-Optimal Call Admission Policies fro Cellular Networks Using Genetic Algorithms, A. Yener & C. Rose, IEEE Wireless'94, Calgary, pp.398–410, July 1994.

Resource Allocation for Wireless Networks, J. MacLellan & C. Rose, VTC'94, Stockholm, pp804–808, June 1994.

Scheduling Arrivals to Queues for Minmax Blocking: the S(n)/M/1/C system, R. Yates & C. Rose, Conference on Information Science and Systems (CISS'94), Princeton, pp.1047–1050, March 1994.

Calculating Cellular Network Performance Using Queueing Methods, J. MacLellan & C. Rose, Conference on Information Science and Systems (CISS'93), Baltimore, pp.101–106, March 1993.

A Dielectric-Free Superconductive Coaxial Cable, C. Rose, AAAS Annual meeting, San Francisco, January, 1990.

A Minimum Distance Automata in Parallel Networks for Optimum Classification, J.H. Winters & C. Rose, INNS Annual Meeting poster session, San Diego, June, 1987.

Invited Papers and Seminars

Interference Avoidance, D. Popescu & C. Rose, Wiley Encylopedia of Telecommunications, J. Proakis, Ed., Wiley, 2002

Paging and Registration in Mobile Networks, C. Rose, Wiley Encylopedia of Telecommunications, J. Proakis, Ed., Wiley, 2002

Keynote Address: *In Wireless Data Networks, Patience Has Its Rewards*, IEEE MWCN, August, 2001, Recife, Brazil

Wireless Data: patience has its rewards, IEEE CCW, October 18, 2000, Isla Captiva FL

Infostations: a new paradigm for wireless data, Nokia Research Center, August 2000 Dallas

MINE MINE MINE: Information Theory, Infostation Networks and Resource Sharing, A. Iacono and C. Rose, WCNC 2000, September 2000, Chicago.

Infostations, Interference Avoidance and the Future of Wireless Data, Georgia Institute of Technology, October 2, 2000

Sum Capacity and Interference Avoidance: convergence via class warfare, C. Rose, CISS 2000, Princeton, March 2000.

Interference Avoidance for Hoards of Things, C. Rose, NASA Glenn Research Center, October, 1999.

Interference Avoidance in Wireless Systems: everything old seems new again, C. Rose, AT&T Newman Springs Seminar, October, 1999.

A Theory of Interference Avoidance, Georgia Tech ECE Seminar, Atlanta, Fall 1998

Optimum Power Schedules for CDMA Access Channels, A. Yener, C. Rose & R. Yates, INFORMS'98, Montreal.

New Wrinkles in Resource Allocation: Infostations and the U-NII, DIMACS Workshop on Networks in Distributed Computing, October, 1997.

Paging and Tandem Queues with Reneging, C. Rose & C. Saraydar, ORSA/INFORMS'97.

Location Uncertainty in Mobile Networks: a theoretical framework, C. Rose & R. Yates IEEE Communications Magazine, 35(2), pp.94–101, Februrary (1997)

Signaling Cost and Location Uncertainty in Mobile Communications Systems: Glimmerings of a Theory ... or Why Information Theory, the Lingua Franca of Uncertainty, Won't Help You Find Mobile Users, Stanford University, June 1996

Polling Queue Disciplines for Paging, ORSA-INFORMS, Spring 1996

Decision-Making in Telecommunications Systems Using Genetic Algorithms, NYNEX, January 1995.

Survival in Graduate School: a perspective from both sides of the fence, Eta Kappa Nu induction banquet address, Rutgers University, November 1994.

Paging and Registration Schemes in Mobile Communications Systems, Rose, C., ITC Miniconference, Ottawa, Quebec Province, Canada, October 1994.

User Location Probability and Paging/Registration Methods in Mobile Mommunications Systems, RUTCOR at Rutgers University, October 1994.

A New Approach to Paging and Registration in Mobile Communications Systems, Bellcore, Middletown, N.J., September 1994.

Paging Registration and Time-Varying User Location Probability, Rose, C., IEEE Miniconference, George Mason University, Fairfax VA, Sept 9, 1994.

Genetic Algorithms and the Search for Good Call Admission Policies: Recent Results, sponsored by D. Mitra, AT&T Bell Laboratories, July 1994.

Genetic Algorithms and the Search for Good Call Admission Policies, RUTCOR at Rutgers University, April 1994.

Wireless Communications Systems: new research challenges, American Physical Society annual meeting, March 1994.

Genetic Algorithms and Call Admission in Telecommunications Networks, ORSA-TIMS, Phoenix, December 1993.

Rapid Optimal Scheduling For TDMA Switches, Princeton University, April 1990.

A Dielectric-Free Superconductive Coaxial Cable, University of Lowell, Lowell Massachusetts, February, 1990.

Rapid Optimal Scheduling For TDMA Switches Using a Cellular Automaton, Brooklyn Polytechnic Institute, Brooklyn, N.Y., April, 1987.

Other Publications

Interference Avoidance and Dispersive Channels: a new look at multicarrier modulation D. Popescu and C. Rose, Winlab Technical Report TR 188, September 1999

Convergence of Greedy Interference Avoidance and Sum Capacity Maximization, C. Rose Winlab Technical Report TR 187 October 1999

Interference Avoidance in Wireless Systems, Christopher Rose, Sennur Ulukus, Roy Yates, Winlab Technical Report TR 175, February 1999,

The Use of Extended Kalman Filter for Emitter Localization in a Multipath Environment, Dimitrie C. Popescu and Christopher Rose, Winlab Technical Report TR 180, April 1999

Minimizing File Delivery Delay in an Infostations Systems, Ana Lucia Iacono and Christopher Rose, Winlab Technical Report TR 167, August 1998

Optimal Routing Schemes for a Simplified Packet Radio Network, Kabir Kasargod and Christopher Rose, Winlab Technical Report TR 164, May 1999

One-Dimensional Location Area Design, Cem U. Saraydar, Owen Kelly, Christopher Rose, Winlab Technical Report TR 161, June 1998,

Selection of Good Location Areas from Population and Traffic Data, C. U. Saraydar and C. Rose, Winlab Technical Report TR 150, May 1997,

Optimum Power Scheduling for CDMA Access Channels, Aylin Yener , Christopher Rose , Roy D. Yates, Winlab Technical Report TR 148, April 1997

Soft Dropping Power Control, Roy Yates, Sorabh Gupta, Christopher Rose, Surgwon Sohn Winlab Technical Report TR 146, March 1997

Wireless Subscriber Location Tracking for Adaptive Mobility Management, Z. Lei & C. Rose, Winlab Technical Report TR 131, (1996).

Toward A Fundamental Theory of Mobility Management, C. Rose & R. Yates, Winlab Technical Report TR 128, (1996).

Minimizing Paging Channel Bandwidth for Cellular Traffic, C.U. Saraydar & C. Rose, Winlab Technical Report TR 125, (1996).

Genetic Algorithms Applied to Cellular Call Admission: Local Policies, A. Yener and C. Rose, Winlab Technical Report TR 112, (1995).

Analysis of a Mobile Assisted Adaptive Location Management Strategy, R. Yates & C. Rose, Winlab Technical Report TR 111, (1995).

Strategies for Page Request Queueing in Mobile Communications Networks, C. Rose & R. Yates, Winlab Technical Report TR 110, (1995).

Highly Mobile Users and Paging: optimal polling strategies, C. Rose & A. Yener, Winlab Technical Report TR 109, (1995).

Near-Optimal Call Admission Policies for Cellular Networks Using Genetic Algorithms, A. Yener & C. Rose, Winlab Technical Report TR 108, (1995).

State-Based Paging/Registration: A Greedy Approach, C. Rose, Winlab Technical Report TR 92, (1995).

Scheduling Arrivals at a Single Server for Min-max Blocking, R. Yates & C. Rose, Winlab Technical Report TR 86 (1994).

Scheduling Arrivals to Queques for Minimum Average Blocking: the S(n)/M/C/C system, C. Rose & R. Yates, Winlab Technical Report TR 85 (1994).

Genetic Algorithms and Call Admission to Telecommunications Networks, C. Rose & R. Yates, Winlab Technical Report TR 84 (1994).

Minimizing the Average Cost of Paging and Registration: a timer-based method, C. Rose, Winlab Technical Report TR 83 (1994) (submitted for publication).

Minimizing the Average Cost of Paging Under Delay Constraints, C. Rose & R. Yates, Winlab Technical Report TR 82 (1994).

Scheduling Arrivals for Min-max Blocking, R. Yates, & C. Rose, (1994).

Analysis and Simulation of Wireless Network Performance Using Queueing Models, J. MacLellan & C. Rose, Winlab Technical Report TR 61 (1993).

Optimal Call Admission to Single Cells of a Cellular Mobile Network, C. Rose & R. Yates, Winlab Technical Report TR 60 (1993).

Mobility Analysis of Call Records in a Cellular Switch, R. Yates & C. Rose, Winlab Technical Report TR 59, (1993).

Grants

($706,729) National Science Foundation CCR 02-05362, *ITR Collaborative Research: Achieving Innovative and Reliable Services in Unlicensed Spectrum*, PI: R. Yates, co-PIs: C. Rose, N. Mandayam, P. Spasojevic, D. Raychaudhuri

($857,000) National Science Foundation CCR 00-85986, *ITR: Collaborative Research: 'Free' Bits: The Challenge of the Wireless Internet.*, PI: R. Yates, co-PIs: C. Rose, N. Mandayam

($430,000) National Science Foundation CCR 99-73012, *Interference Avoidance in Wireless Systems*, PI: C. Rose, co-PI: R.D. Yates

($5,000) National Science Foundation CCR 98-14104, *A Workshop to Study Peaceful Coexistence in the Unlicensed National Information Infrastructure*, PI: C. Rose, co-PI: A. Ogielski, D. Goodman

($1,250,000) New Jersey Commission on Science and Technology, *New Jersey Center for Wireless Technology (99-2042-007-17)*, PIs: R. Frenkiel, N. Mandayam, C. Rose, R. Yates.

($170,000) National Science Foundation NCR 97-29863 *Parallel Computing for Wireless Networking Research*, PI: D.J. Goodman co-PIs: N.B. Mandayam, A.T. Ogielski, C. Rose, R.D. Yates

($20,000) *Genetic Algorithms and the Control of Wireless Telecommunications Systems*, AT&T Laboratory Improvement Grant, Fall, 1995.

($469,897) National Science Foundation 95-06505 NCRI, 1/96–1/99 *Power Control and Packet Radio Networks*, PI: R. Yates, co-PI: C. Rose.

($412,456) National Science Foundation 92-06148 NCRI, 9/92–9/95, *Searching for Good Call Admission Policies in Telecommunications Networks*, PI: C. Rose, co-PI: R. Yates.

($100,000) Startup funds from Rutgers University, 9/1/90

($15,000) Henry Rutgers Research Fellowship 9/1/90

($50,000) Cummulative equipment donations from AT&T Bell Laboratories

Grants Pending

($260,969) National Science Foundation CCR-0218963, *ITR Collaborative Research: Interference Avoidance and Peaceful Coexistence in Wireless Systems*, PI: C. Rose

Graduate Students

Completed Doctoral Students

- John MacLellan (1994) *Methods of Resource Allocation in Cellular Networks Using An Underlying Queueing Model*
- Ana-Lucia Iacono (2000) *Information Delivery in an Infostation Network*
- Dimitrie Popescu (2002) *Interference Avoidance for Wireless Systems*

Current Doctoral Students

- Otilia Popescu *Interference Avoidance for Multiple Base Systems*
- Jasvinder Singh *Codeword Feedback for Interference Avoidance*

Completed Masters Students

- Aylin Yener (1994) *Genetic Algorithms and Call Admission in Cellular Systems*,
- Zhuyu Lei (1996) *Wireless Subscriber Location Tracking for Adaptive Mobility Management*,
- Kenyon Wells (1997) *A Hardware Testbed for Genetic Algorithms*
- Cem Saraydar (1997) *Deriving Optimal Location Areas from Measured Vehicular Traffic*
- Kabir Kasargod (1999) *Routing in Simple Packet Radio Networks*
- David Tabora (2001) *Covariance Estimation, Feedback Channel Analysis and Multiple Base Performance of Interference Avoidance*

Currently Supervised Masters Students

- Furuzan Atay, *Demon-Net: an ad-hoc network based on mobility*
- Pavan Kaivaram, *Methods for High Speed High SNR Channels*

Visiting Scholars

- David Cavalcanti, Universidade Federal de Pernambuco *A Mobile Infostation System*

Teaching

Instructor: *14:330:450 Principles of Communications*, Rutgers University

Typical enrollment: 100

An introduction to analog modulation techniques. Increased emphasis on orthogonal signal representations and introduced elementary detection theory for easier transition to 330:452 (see below). TEXT: Stremler, Introduction to Communications Systems *(Fall 1990-1993, Spring 1992)*

Instructor: *14:330:452/14:332:421 Communications Engineering*, Rutgers University

Typical enrollment: 40

Rigorous introduction to digital communications, signal space, quantization and information theory.
TEXTS: Proakis & Salehi, Communications Systems Engineering (1995); Haykin, Communication Systems (1996)

Instructor: *16:332:426 Wireless PCS*, Rutgers University

Typical enrollment: 10

Undergraduate introduction to wireless systems. Have updated the course from a compendium of specific methods used for specific systems to a higher level theoretical treatment applicable to all wireless mobile networks. Introduced a final project where students design a wireless mobile network for services of their own design
TEXTS USED: Ahlin & Zander, Principles of Wireless Communications (2002); Stuber, Principles of Wireless Communications (2001); Goodman, Wireless Personal Communications Systems (1999-2000)

Instructor: *16:332:501 Systems Analysis*, Rutgers University

Typical enrollment: 50

Graduate linear systems course. Graduate level introduction to systems theory with special emphasis on measure-theoretic concepts for broader applicability of results. State-space, transform theory, orthogonal projections and elementary integral equations.
TEXT: Szidarovsky & Bahill, Linear Systems (1995–1998)

Instructor: *16:330:541 Stochastic Signals and Systems*, Rutgers University

Typical enrollment: 35

Graduate probability and stochastic processes course. Greatly increased emphasis on stochastic processes and estimation.
TEXT: Papoulis, Probability, Random Variables and Stochastic Processes (1994,1999); Yates & Goodman, Probability and Stochastic Processes (2000–2001)

Instructor: *16:332:542 Information Theory and Coding*, Rutgers University

Typical enrollment: 10

Graduate introduction to information theory, with heavier emphasis on broad applicability of IT concepts (portfolio management, location tracking, estimation).
TEXTS: Blahut, Principles and Practice of Information Theory (1991); Cover & Thomas, Elements of Information Theory (1992-1996)

Instructor: *16:332:545 Communications Theory*, Rutgers University

Typical enrollment: 20

Graduate introduction to communication theory. Optimal detection, optimal receivers, signal space, modulation schemes, brief introduction to information theory and coding.
TEXT: Proakis, Communications Systems Engineering; class notes (1999)

Supervisor: *14:330:491 Undergraduate Special Problems*, Rutgers University

Jose Almeda	*Transient Properties of Voice Traffic Modeled by Birth and Death Processes*	Spring 1991
Andrea Fogel	*Sudden Infant Death Apnea Monitor: a design study*	Spring 1991
Mark Rosen	*Queueing Network Simulator with Graphical Front End*	Spring 1991
Fotios Papayianis	*Television Signal Analysis: internship project report*	Fall 1992
Michele Mekjian	*A Control System for Scanning Tunneling Microscopy*	Spring 1993
Walead Mosaad	*Mobile System Communication*	Spring 1994
Patrick Keane	*Security and the Wireless Checkbook: application of cryptographic methods*	Fall 1995
Tamyra Holman	*Parity Codes for Memory Chips*	Fall 1997
Charles Kim	*CNN.com and Yahoo.com Website Update Information*	Fall 2001
Murtaza Manaqibwala	*Simple Signaling for High SNR Channels*	Spring 2002

University Activities

University Committee on Recruitment Policy

ECE Ad hoc PhD. Preliminary Examination Review Committee

ECE Scholastic Standing Committee

ECE Teaching Assignment Committee

ECE External Review Committee

Professional Activities

Keynote Address: *In Wireless Data Networks, Patience Has Its Rewards*, IEEE MWCN 2001, Recife, Brazil, August 2001

Tutorial Instructor: *Wireless Communications 100 + New Paradigms*, IEEE MWCN 2001, Recife, Brazil, August 2001

Conference General Chair: ACM MobiCom 2001, Rome, July 2001

Conference Co-Chair: Berkeley/WINLAB Focus 2000 on PicoRadio Networks, Berkeley (June, 2000)

International Visiting Committee: Portuguese Foundation on Science and Technology nation-wide engineering program assessment, (December, 1999).

Conference Co-Chair: WINLAB/Berkeley Focus'99 on Radio Networks for Everything, New Brunswick, NJ (May, 1999)

Conference Co-Chair: WINLAB Focus'98 on the U-NII, Long Branch, NJ (June, 1998).

Technical Program Committee Co-Chair: MobiCom'97, Budapest, September, 1997.

New Jersey Commission on Science and Technology: Scientific Fields Committee Advisory Group.

Consultant: Wireless Communications (industry/venture capital/legal)

Executive Committee: ACM SIGMobile (1997-2000)

Steering Committee: ACM MobiCom, ACM SIGMobile WoWMoM

Editor: IEEE Transactions on Vehicular Technology

Editor: Elsevier Computer Networks

Editor: ACM Wireless Networks

Guest Editor: ACM MONET, Special issue on mobility, November, 1996, ACM WINET, Selected Papers from Mobicom'97, December, 1999).

Technical Committee Membership:

- IEEE Infocom 2003
- IEEE Globecom: Internet Performance Symposium 2002
- ACM MobiCom: Wireless Sensor Networks Workshop 2002
- IEEE MCWN 2001
- ACM MobiCom 2000

- IEEE MMT'98
- ACM MobiCom'98
- IEEE Infocom'98
- IEEE MMT'97
- ACM MobiCom'97
- ACM MobiCom'96

Panelist

- National Science Foundation, Arlington, Va.
- NRI/Ford Foundation Fellowship Selection Committee

Session Chairman

- Modeling and Identification of Fading Channels, CISS 2000, Princeton, March 2000
- QoS in Wireless Networks, ACM WoWMoM'99, San Francisco, August 1999
- Mobility Management, Infocom'98, San Francisco, April 1998
- Mobile Paging and Tracking, ICUPC'96, Boston MA, October 1996
- Mobile Data, IEEE VTC, Stockholm, Sweden, June 1994

Member: Institute of Electrical and Electronic Engineers
Association for Computing Machinery
New York Academy of Sciences
American Association for the Advancement of Science